

3.7 2020 量物

(1)  $E_n = \frac{e^2}{2an^2}$ , 简并度  $2n^2$  (算上电自旋)

(2)

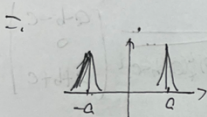
① 势 (1D 势), 由电势  $\phi(x)$  求得  
 势  $\phi(x) = \frac{e}{4\pi\epsilon_0} \int \frac{\rho(x')}{|x-x'|} dx'$   
 和电势  $\phi(x)$  关系, 如 (1D, 1D 势)

② 求势  $\phi(x)$

(3)

$E_n = \frac{e^2}{2an^2}$ ,  $a = \frac{4\pi\epsilon_0\hbar^2}{m}$

$= \frac{e^2}{2 \cdot 6.072 \cdot 10^{-10} \text{m}}$



(4)

① 在  $x = -a$  处

$\psi(-a) - \psi'(a) = \frac{2\hbar^2}{m} \psi(a)$ ,  $\psi(-a) = \psi(a)$

② 在  $x = a$  处,  $\psi'(a) - \psi(-a) = \frac{2\hbar^2}{m} \psi(a)$ ,  $\psi(a) = \psi(-a)$

(5)

① 在  $x = -a$  处

$Bk_1 e^{-k_1 a} - k_1 \sinh k_1 a = \frac{2\hbar^2}{m} B e^{-k_1 a}$  ①  
 $B e^{-k_1 a} = \cosh k_1 a$  ②

② 在  $x = a$  处

$k_2 \cosh k_2 a = C k_2 e^{-k_2 a} = \frac{2\hbar^2}{m} \sinh k_2 a$   
 $\sinh k_2 a = e^{-k_2 a}$

(6)

$k_1 a = 1 - k_1 a$

② 在  $x = 0$  处

$k_1 \cosh k_1 a + k_1 \sinh k_1 a = \frac{2\hbar^2}{m} B e^{-k_1 a}$

$k_1 e^{k_1 a} = \frac{2\hbar^2}{m} B e^{-k_1 a}$

$B = \frac{\hbar^2}{2\hbar^2} k_1 e^{2k_1 a}$

同样在  $x = a$  处

$C k_2 e^{-k_2 a} - k_2 \cosh k_2 a = \frac{2\hbar^2}{m} C e^{-k_2 a}$  ①

$C e^{-k_2 a} = \sinh k_2 a$  ②

③ 在  $x = 0$  处

$(-k_1) e^{k_1 a} = \frac{2\hbar^2}{m} C e^{-k_1 a}$

$C = (-k_1) \frac{\hbar^2}{2\hbar^2} e^{2k_1 a}$

(7)

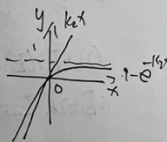
(4)  $k_1 a = 1 - k_1 a$

① 在  $x = 0$  处

② 在  $x = a$  处

当  $k_1 = k_2$  时

否则有解



三. 2.1.

(1) 选取基矢:

$$\vec{S}_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\vec{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{H} = A\vec{S}_z + B\vec{S}_y^2$$

$$= A\hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} + B\frac{\hbar^2}{4} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{A\hbar}{2} + \frac{B\hbar^2}{2} & 0 & \frac{B\hbar^2}{2} \\ 0 & A\hbar & 0 \\ \frac{B\hbar^2}{2} & 0 & \frac{A\hbar}{2} - \frac{B\hbar^2}{2} \end{pmatrix} \quad \text{设 } A = A\hbar, B = \frac{B\hbar^2}{2}$$

$$= \begin{pmatrix} a+b & 0 & b \\ 0 & 2b & 0 \\ b & 0 & b-a \end{pmatrix}$$

$$\begin{pmatrix} a+b & 0 & b \\ 0 & 2b & 0 \\ b & 0 & b-a \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix}$$

$$\begin{vmatrix} a+b-E & 0 & b \\ 0 & 2b-E & 0 \\ b & 0 & b-a-E \end{vmatrix} = 0$$

$$(2b-E)(E^2 - 2bE - a^2) = 0$$

$$\therefore E_1 = 2b, E_2 = b + \sqrt{a^2 + b^2}$$

$$E_3 = b - \sqrt{a^2 + b^2}$$

$$\text{即 } E_1 = \hbar^2 B$$

$$E_2 = \frac{\hbar^2 B^2 + \sqrt{4A^2 \hbar^2 + B^2 \hbar^4}}{2}$$

$$E_3 = \frac{B\hbar^2 - \sqrt{4A^2 \hbar^2 + B^2 \hbar^4}}{2}$$

$$(2) |\psi(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle$$

由(1)解得  $\hat{H}$  的本征矢为

$$|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, |2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$|2\rangle = \frac{1}{\sqrt{(a-b+c)^2 + (a+b-c)^2}} \begin{pmatrix} a-b+c \\ 0 \\ a+b-c \end{pmatrix}$$

$$|3\rangle = \frac{1}{\sqrt{(a+b-c)^2 + (a-b-c)^2}} \begin{pmatrix} a-b-c \\ 0 \\ a+b+c \end{pmatrix}$$

$$\therefore |\psi(t)\rangle = \sum_n \langle n | \psi(0) \rangle e^{-iE_n t/\hbar} |n\rangle$$

$$= \frac{1}{2} \sum_n C_n |n\rangle$$

$$|\psi(t)\rangle = C_1 e^{-iE_1 t/\hbar} |1\rangle + C_2 e^{-iE_2 t/\hbar} |2\rangle + C_3 e^{-iE_3 t/\hbar} |3\rangle$$

$$\text{其中 } C_1 = \frac{a-b+c}{\sqrt{(a-b+c)^2 + (a+b-c)^2}}, C_2 = \frac{a-b-c}{\sqrt{(a-b-c)^2 + (a+b+c)^2}}, C_3 = \frac{a+b-c}{\sqrt{(a-b-c)^2 + (a+b+c)^2}}$$

$$P_{2,1} = |\langle 2 | \psi(t) \rangle|^2, |\psi\rangle = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{四. } \psi(x) = A e^{-\frac{\alpha x^2}{2}}$$

(1)

$$1 = \int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx$$

$$= A^2 \int_{-\infty}^{\infty} e^{-\alpha x^2} dx$$

$$= 2A^2 \int_0^{\infty} e^{-\alpha x^2} dx$$

$$= 2A^2 \cdot \frac{\sqrt{\pi}}{2\alpha}$$

$$\therefore A = \sqrt{\frac{\alpha}{\pi}}$$

$$\therefore \psi(x) = \sqrt{\frac{\alpha}{\pi}} e^{-\frac{\alpha x^2}{2}}$$

$$P(x) = |\psi(x)|^2$$

$$= \frac{\alpha}{\pi} e^{-\alpha x^2}$$

(2)

$$\langle x \rangle = \int_{-\infty}^{\infty} x \psi^*(x) \psi(x) dx$$

$$= 0$$

$$\langle x^2 \rangle = \langle 0 | x^2 | 0 \rangle$$

$$= 0$$

$$\langle x^2 \rangle = \langle 0 | x^2 | 0 \rangle$$

$$= \frac{1}{2\alpha^2}$$

$$\langle p \rangle = \langle 0 | p | 0 \rangle$$

$$= 0 - \frac{\hbar}{i} \langle 0 | \frac{\partial}{\partial x} | 0 \rangle$$

$$= 0$$

$$\langle p^2 \rangle = \langle 0 | p^2 | 0 \rangle$$

$$= \frac{\hbar^2 \alpha^2}{2}$$

(3) 能量本征态为  $N$  个,  $\hat{H} = (\hat{H}_0 + \hat{H}_1)$  故

$$\hat{H}_0$$
 能量本征态为  $|0\rangle$

(4)  $\langle \Delta x \rangle \langle \Delta p \rangle \geq \frac{\hbar^2}{4}$

$$\langle \Delta x \rangle = \Delta x - \bar{x}$$

$$= \frac{\hbar^2}{2\alpha^2}$$

$$\langle \Delta p \rangle = \bar{p} - p$$

$$= \frac{\hbar^2 \alpha^2}{2}$$

$$\therefore \langle \Delta x \rangle \langle \Delta p \rangle = \frac{\hbar^2}{4}$$

$$\langle \Delta x \rangle \langle \Delta p \rangle \geq \frac{\hbar^2}{4}$$

(5)

$$a = \sqrt{\frac{\hbar \omega}{2m}} (x + \frac{i}{\hbar \omega} p)$$

$$a^\dagger = \sqrt{\frac{\hbar \omega}{2m}} (x - \frac{i}{\hbar \omega} p)$$

$$\therefore x = \frac{1}{\sqrt{2} \sqrt{\frac{\hbar \omega}{m}}} (a + a^\dagger)$$

$$\hat{p} = i \sqrt{\frac{\hbar \omega}{2}} (a^\dagger - a)$$

在  $\hat{H}_0$  基下:

$$\hat{H}_0 = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}$$

$$\hat{H} = \hat{H}_0 + \hat{H}_1$$

(6) 由于  $E_1 \neq E_2$  为非简并微扰

$$\therefore E_1' = E_1 + \Delta E_1 = 2$$

$$E_1^{(1)} = \sum_{m \neq 1} \frac{|H_{m1}|^2}{E_1^{(0)} - E_m^{(0)}} = \frac{|\beta_2|^2}{E_1^{(0)} - E_2^{(0)}}$$

$$|\psi_1^{(1)}\rangle = \sum_{m \neq 1} \frac{H_{m1}}{E_1^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle = \frac{\beta_2}{E_1^{(0)} - E_2^{(0)}} |\psi_2^{(0)}\rangle$$

$$\therefore E_1' = E_1 + \alpha + \frac{|\beta_2|^2}{E_1^{(0)} - E_2^{(0)}}$$

$$|\psi_1'\rangle = |\psi_1^{(0)}\rangle + \frac{\beta_2}{E_1^{(0)} - E_2^{(0)}} |\psi_2^{(0)}\rangle$$

同理对于  $E_1'$

$$\text{有: } E_1^{(0)} = E_1, E_1^{(1)} = H_{21}' = 2$$

$$E_2^{(1)} = \sum_{m \neq 1, 2} \frac{|H_{m1}'|^2}{E_2^{(0)} - E_m^{(0)}} = \frac{|\beta_1|^2}{E_1^{(0)} - E_2^{(0)}}$$

$$|\psi_1^{(1)}\rangle = \frac{\beta_1}{E_1^{(0)} - E_2^{(0)}} |\psi_2^{(0)}\rangle$$

$$|\psi_2^{(1)}\rangle = \sum_{m \neq 1, 2} \frac{H_{m2}'}{E_2^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle$$

$$= \frac{\beta_1}{E_1^{(0)} - E_2^{(0)}} |\psi_1^{(0)}\rangle$$

$$\therefore |\psi_2^{(1)}\rangle = |\psi_2^{(0)}\rangle + \frac{\beta_1}{E_1^{(0)} - E_2^{(0)}} |\psi_1^{(0)}\rangle$$

$$E_2' = E_2 + 2 + \frac{|\beta_1|^2}{E_2^{(0)} - E_1}$$

2.  $\beta_1 = \beta_{12}, E_1 = E_2$  为简并态

$$\therefore H' = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\text{且 } |\psi^{(0)}\rangle = C_1 |\psi_1\rangle + C_2 |\psi_2\rangle$$

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E^{(0)} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\text{显然 } E^{(1)} = E_1^{(0)} = 0$$

因此  $E_1, E_2$  仍为简并态

$$\text{故 } |\psi^{(0)}\rangle = C_1 |\psi_1\rangle + C_2 |\psi_2\rangle, |\psi^{(1)}\rangle = \sum_{m \neq 1, 2} \frac{H_{m1}'}{E_1^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle$$

= 0 (无修正)

$$H_0 |\psi_n^{(0)}\rangle + H' |\psi_n^{(0)}\rangle = E^{(0)} |\psi_n^{(0)}\rangle + E^{(1)} |\psi_n^{(0)}\rangle + E^{(2)} |\psi_n^{(0)}\rangle$$

$$(H_0 - E^{(0)}) |\psi_n^{(0)}\rangle = E^{(2)} |\psi_n^{(0)}\rangle + (E^{(1)} - H') |\psi_n^{(0)}\rangle$$

两边左乘  $\langle \psi_1 |$  得

$$\beta_1 \int C_1 E^{(2)} = \langle \psi_1 | H' | \psi^{(0)} \rangle$$

$$\int C_2 E^{(2)} = \langle \psi_2 | H' | \psi^{(0)} \rangle$$

$$|\psi^{(0)}\rangle = \sum_{m \neq 1, 2} \frac{H_{m1}'}{E_1^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle$$

$$= \sum_{m \neq 1, 2} \frac{\langle \psi_m^{(0)} | H' | C_1 |\psi_1\rangle + C_2 |\psi_2\rangle \rangle}{E_1^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle$$

$$= \sum_{m \neq 1, 2} C_1 \frac{H_{m1}' H_{m1}^{(0)}}{E_1^{(0)} - E_m^{(0)}} + C_2 \sum_{m \neq 1, 2} \frac{H_{m2}' H_{m1}^{(0)}}{E_1^{(0)} - E_m^{(0)}} |\psi_m^{(0)}\rangle$$

$$\therefore C_1 E^{(2)} = C_1 \sum_{m \neq 1, 2} \frac{H_{m1}' H_{m1}^{(0)}}{E_1^{(0)} - E_m^{(0)}} + C_2 \sum_{m \neq 1, 2} \frac{H_{m2}' H_{m1}^{(0)}}{E_1^{(0)} - E_m^{(0)}}$$

$$C_2 E^{(2)} = C_1 \sum_{m \neq 1, 2} \frac{H_{m2}' H_{m1}^{(0)}}{E_1^{(0)} - E_m^{(0)}} + C_2 \sum_{m \neq 1, 2} \frac{H_{m2}' H_{m2}^{(0)}}{E_1^{(0)} - E_m^{(0)}}$$

$$\begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E^{(2)} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$